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B.Tech. Degree VII Semester Examination November 2014

EB/EC/EI 705 (D) MECHATRONICS (2006 Scheme)

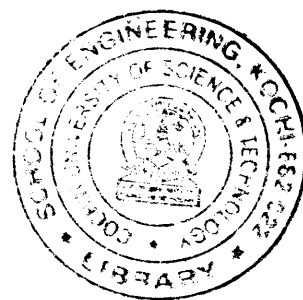
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain the principles of position and proximity measurements.
- (b) Explain the role of filters and signal conditioners in data acquisition stems.
- (c) Explain the working of hydraulic pumps, with suitable diagram.
- (d) What is a solid state relay? Compare with mechanical relay.
- (e) Explain the internal structure of MEMS. List the applications.
- (f) What is mechatronics? How is it useful in robotic applications?
- (g) Draw the characteristics of ON-OFF controller and explain.
- (h) Explain the ladder programming of PLC with an example.



PART B

(4 × 15 = 60)

- II. (a) Draw the block diagram of mechatronics system and explain the components involved. (8)
- (b) Explain the process of signal measurement and calibration. (7)
- OR**
- III. (a) Draw the block diagram of data acquisition system and explain. (8)
- (b) Explain with diagrams, the sensory system for temperature, force and velocity. (7)
- IV. (a) Give the structure of process control valve and explain. Write a note on valve size and materials. (8)
- (b) Write short notes on cams and gear trains in mechatronics. (7)
- OR**
- V. (a) What is an actuator? Explain linear and rotary types, with suitable sketches. (8)
- (b) Explain the working principles of servo and stepper motors. (7)
- VI. (a) Develop the mathematical model of a typical rotational system and explain. (9)
- (b) Explain the principle of opto isolator. How is it useful in mechatronic systems? (6)
- OR**
- VII. (a) Draw a typical electro-mechanical system. How is its step response analysed? (9)
- (b) Explain the manufacturing process of MEMS. (6)
- IX. (a) Draw the circuit/schematic of PID controller. Derive its transfer function. Compare with PI control. (10)
- (b) Write short note on FMS. (5)
- OR**
- X. (a) Draw the schematic of a CNC machine and explain. Where is it preferred over other control schemes? Discuss with an example. (9)
- (b) Explain the principle of intelligent control. (6)